

Response to Reviewers

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Discover Artificial Intelligence

Collection: Trustworthy and Responsible Federated Learning

Revised title	Unifying Fairness Privacy Robustness and Explainability in Federated Graph Neural Networks for High Stakes Risk Detection
Previous title	Trustworthy Federated Graph Neural Networks: Unifying Fairness, Privacy, Robustness, and Explainability for High-Stakes Risk Detection
Decision	Minor revision
Enclosed	Clean revised manuscript; tracked-changes manuscript (all edits marked); this response letter

Dear Dr. Fisichella and Ms. Baig, and dear Reviewers,

We thank the Editor and both Reviewers for a careful, constructive and unusually useful set of comments. We are grateful that the reviewers found the work technically sound and well motivated, and we have taken the criticisms seriously: **every point raised has been addressed**, and in several cases we went further than asked.

Following the Editor’s instruction to check the consistency between the text and the reported tables, we re-verified *every* numerical claim in the paper against the table it cites. Text and tables now agree throughout.

In this letter, reviewer text appears in *grey boxes*, our response follows in plain text, and new or rewritten manuscript text appears in *green-bordered boxes*. Page numbers refer to the clean revised manuscript. Nothing in the revision required new experiments: as Reviewer 1 anticipated, all changes concern wording, scope and consistency. One figure and two table columns were regenerated from the *existing* logged results, with no re-training; we note each below.

1. Summary of changes

Raised by	Issue	Where fixed
Editor	Title contains punctuation	New title (title page)
Editor	Declarations need separate subheadings	Declarations, p. 38
Editor	Data availability statement	Declarations, p. 38
Editor/R1	Overstated claims	§5.8 p. 25, §5.3 p. 18, §6 p. 31
Editor/R1	Text/table consistency	§5.3, §5.4, §5.6, §5.8
Editor/R1	Scope of the DP guarantee	Abstract, §1, Table 1 note, §4.2
Editor/R2	Protected attribute at inference	New paragraph, §6 p. 32
Editor/R2	Trust-score limitations	§6 p. 31 (moved before the score)
R1.1	<code>robust_bfwa</code> exclusivity claim	§5.8 p. 25, §5.4 p. 20
R1.2	Sixteen vs. fifteen baselines	Throughout; §5.2 p. 15
R1.3	Uneven baseline coverage	New paragraph, §5.2 p. 17
R1.4	Single-seed robustness	New paragraph §5.8 p. 26; Limitation (4)
R1.5	Energy column reads 0.00 Wh	Table 11 and Table 10, §5.10
R1.6	Five text/table mismatches	§5.3, §5.4, §5.6
R1.7	DP scope in the abstract	Abstract; Table 1 note
R2.1	Inference-time s needs a full treatment	New paragraph §6 p. 32; Limitation (12) p. 36
R2.2	Trust-score caveat comes too late	§6 p. 31; Limitation (10)

2. Response to the Editor

E1. Revised title

Revise Title: To aid our readers, and to maximize the accessibility of your manuscript, the title should have a clear, precise scientific meaning. Where possible, the title should be read as one concise sentence, without the use of punctuation (full stops, colons, hyphens, question marks etc.). Please could you re-write the title ensuring that it is informative and appropriate?

Done. The previous title used a colon, three commas and a hyphen. The revised title contains **no punctuation of any kind**, reads as a single sentence, and still names all four trust properties and the application domain:

Unifying Fairness Privacy Robustness and Explainability in Federated Graph Neural Networks for High Stakes Risk Detection

We removed the hyphen from “High-Stakes” throughout the title and kept a short running head (“Trustworthy Federated Graph Neural Networks”) for the page footer, as the journal template requires.

E2. Declarations section with individual subheadings

Authors are requested to include a separate section in the manuscript titled “Declarations.” In this section, please clearly state the following items as individual subheadings: Ethical approval; Consent to participate; Consent to publish. Each item should be addressed separately

and appropriately.

Done [Declarations, p. 38]. The Declarations were previously a single bulleted list. They are now a section in which *each* item carries its own subheading: Funding; Conflict of interest; **Ethical approval**; **Consent to participate**; **Consent to publish**; Data availability; Materials availability; Code availability; Author contributions. Each is addressed substantively rather than with a bare “not applicable” — for example:

Ethical approval. Not applicable. This study involved no human participants and no animal subjects. It uses only publicly released, de-identified benchmark datasets (German Credit, Credit Defaulter, Bail/Recidivism, Pokec, Elliptic, and ogbn-products), each obtained from its published distribution and used in accordance with its licence and intended research purpose. No ethics-committee approval was therefore required.

E3. Data availability statement

When submitting your revised manuscript, please select ‘yes’ under the data availability declaration ... Please make sure that data availability statement in the manuscript and system exactly matches.

Done [Declarations, p. 38]. The manuscript now contains an explicit **Data availability** subheading naming every dataset and its public source with URLs, and stating that no new data were generated. We will select “yes” in the submission system and paste this statement verbatim so that the two match exactly.

We also strengthened the **Code availability** statement. The previous wording (“available from the corresponding author upon request ... will be released publicly upon acceptance”) sat awkwardly beside the paper’s reproducibility claims. The repository is now public and cited directly: <https://github.com/vinhqdang/FedFairGNN>. It contains the source code, the serialised configuration defining every reported run, and the generated tables and figures; since every dataset is public and every run is fully specified by its configuration, any reported number can be reproduced end to end from the repository alone.

E4. Overstated claims, text/table consistency, DP scope

For the revision, please address the few overstated or unsupported claims, carefully check the consistency between the text and reported tables, and clarify the scope of the differential privacy guarantee.

These three requests coincide with Reviewer 1’s comments 1, 6 and 7, and we address them in full there (Sections 3, 3, 3 of this letter). In summary:

- **Overstated claims.** The `robust_bfwa` exclusivity claim is withdrawn and replaced with an accurate two-aggregator statement (§5.8), and the claim about non-private baselines on German AUC now names the exception (§5.3); both are detailed under R1.1 and R1.6(c) below. We also went through the paper’s remaining comparative statements and bounded each to what the corresponding table supports.
- **Text/table consistency.** Every numerical claim in the paper has been re-verified against the table it cites; text and tables now agree throughout.
- **DP scope.** The abstract now states the scope limitation in its own sentence rather than in a parenthesis, and the positioning table (Table 1) now carries a footnote so that our ✓ in the DP column cannot be read as equivalent to DP-FedAvg’s record-level guarantee.

E5. Practical implications of requiring protected attributes at inference; trust-score limitations

We also encourage you to expand the discussion of the practical implications of requiring protected attributes at inference time and to make the limitations of the composite trust score more explicit.

Both done, and both expanded well beyond a sentence, since Reviewer 2 asked for the same two things. See Sections 4 and 4 of this letter: a new dedicated paragraph in §6 (p. 32) analysing the lawful basis, data-access, collection and purpose-limitation consequences for a deploying institution, and a restructuring of §6 so the trust score’s caveats are stated *before* the score is presented.

3. Response to Reviewer 1

We thank Reviewer 1 for an exceptionally precise reading. The five text/table mismatches in comment 6 were all real, and finding them required checking our numbers against our own tables line by line; we are grateful for the effort.

R1.1 Softening the `robust_bfw` claim

Section 5.8. The statement that robust-bfw is the only aggregator that stays strong under all three attacks is stronger than what Table 9 shows. The trimmed-mean aggregator achieves comparable utility and even performs better under the ALIE attack. I suggest softening this claim.

The reviewer is correct and we have withdrawn the claim. Trimmed mean retains 0.984/0.956/0.983 AUC across the three attacks against our 0.992/0.953/0.990, so it survives the suite as we do and is indeed marginally better under ALIE. The passage now reads [§5.8, p. 25]:

“... FedAvg and plain BFWA collapse under the Gaussian attack (0.51 and 0.48 AUC) and Krum collapses under ALIE (0.71), whereas `robust_bfw` (0.992/0.953/0.990) and the trimmed mean (0.984/0.956/0.983) both stay strong under all three. We are careful here not to overstate: the table does not support an exclusivity claim for our aggregator, and the trimmed mean in fact retains marginally more utility than ours under ALIE (0.956 vs. 0.953). The accurate statement is that two of the seven aggregators survive the full suite — ours and the trimmed mean — and that among those two, ours is the one that also carries the fairness constraint of Section 4.3. ... The honest summary is that `robust_bfw` achieves a strong overall *balance* ... rather than dominating any metric.”

We have applied the same correction to the parallel statement in the ablation discussion [§5.4, p. 20], so the two sections now agree. The revised passage also names the remaining cells in which a competitor leads: trimmed mean attains the lowest EOD under fairness poisoning (0.005) and multi-Krum the lowest EOD under Gaussian corruption (0.017).

R1.2 Sixteen versus fifteen baselines

Number of baselines. The manuscript refers to sixteen baselines, whereas the main comparison tables appear to contain fifteen. Please check whether one baseline is missing from the tables or revise the reported number throughout the manuscript.

The reviewer is right: the tables contain fifteen baselines and the count of sixteen was wrong. The discrepancy is **FedFB**, which we described among the compared methods but never ran,

because its FairBatch-style local group reweighting coincides in our pipeline with the local soft-demographic-parity objective already carried by FairFed and F²GNN — running it would have duplicated an existing row rather than covering a new axis.

We have corrected the count to **fifteen** in all six places where it appeared (abstract, §1 contribution 5, §1 relation-to-preliminary-version, §5.2, §7 conclusion), removed FedFB from the list of *evaluated* baselines in §5.2 and §2, and retained it only in the related-work survey, where it belongs as literature. To prevent the ambiguity recurring, §5.2 now ties the count directly to the tables [§5.2, p. 15]:

“The fifteen correspond exactly to the fifteen baseline rows of Tables 4–6, above the rule that separates them from our own three variants; no method is described here that does not appear there. (FedFB, surveyed in Section 2, is *not* among them: its FairBatch-style local group reweighting coincides with the local soft-DPD objective already carried by FairFed and F²GNN in our pipeline, so running it would have duplicated an existing row rather than added an axis.)”

R1.3 Uneven baseline coverage across datasets

Baseline comparisons across datasets. The paper gives the impression that all datasets are evaluated against the full set of baselines. However, the Credit and especially the Elliptic datasets include fewer competing methods. This should be stated explicitly in the experimental setup.

Agreed — this was left to be inferred from dashes in the tables, which is not good enough. The experimental setup now carries a dedicated paragraph stating the coverage exactly [§5.2, p. 17]:

“**Baseline coverage is not uniform across datasets.** ... The *full* fifteen-baseline comparison is run only on the three datasets that combine a genuine protected attribute with a tractable cost: **German**, **Bail**, and **Pokec**. **Credit** (30,000 nodes) is run against the seven core baselines ... plus DP-FedAvg, but not against the five 2025–2026 reimplementations or FaVGNN/FDP-Fair. **Elliptic** (203,769 nodes) is run against only FedAvg-GAT and FairSIN, the two baselines that scale to it within our CPU budget ... Consequently the Credit and Elliptic columns of Tables 4–6 should be read as comparisons against *that* restricted set, and a ‘–’ always means *not run*, never a failed or omitted result.”

The consequence is also propagated to the places where those two datasets are discussed: the Elliptic paragraph in §5.3 now says explicitly that only two baselines were run at that scale, so the result is “not a broad competitive ranking”; the introduction flags the restriction when it announces the evaluation; and a new limitation item (4b) records it.

R1.4 Single-seed robustness experiments

Robustness experiments. Since robustness is one of the main contributions, the conclusions are currently based on single-seed experiments. Adding a few additional runs would strengthen the evidence. If this is not feasible, the claims should be slightly moderated.

We have taken the second option and moderated the claims, and we want to be transparent about why. A multi-seed replication of the 7×3 aggregator–attack grid is a substantial compute commitment that we could not complete within the revision window without compromising the rest of the response; we would rather state the limit of the present evidence precisely than add two seeds and still be unable to support a ranking.

What we have done instead is to bound the claims to what one seed can carry. A new paragraph [§5.8, p. 26] separates the order-of-magnitude findings we rely on from the small differences we explicitly decline to interpret:

“How far this single-seed evidence reaches. These are *single-seed* runs on Bail, and we deliberately do not lean on them harder than that allows. Differences of a few thousandths in AUC or EOD between the surviving aggregators — ours, the trimmed mean, multi-Krum — are well within the range seed noise alone produces elsewhere in this paper (TrustFedGNN’s own Bail DPD has a standard deviation of 0.013 over five seeds), so we make no ranking claim among them, and no conclusion in this paper rests on such a gap. What a single seed *does* support, because these effects are order-of-magnitude rather than marginal, are the two qualitative findings we actually rely on: that several aggregators *collapse* outright under a given attack (FedAvg and plain BFWA to ≈ 0.5 AUC under Gaussian noise, Krum to 0.71 under ALIE), and that plain BFWA’s fairness is captured by the fairness-poisoning attack — an EOD of 0.126, more than five times the 0.024 of the screened variant. Replicating the full 7×3 aggregator–attack grid over multiple seeds, with error bars and with adaptive attackers who know the screening rule, is the natural next step ...”

The limitations section was rewritten to match, and now names the robustness study as the place where statistical power runs out and identifies its multi-seed replication as “the most valuable single addition to this evaluation” [§7, Limitation (4), p. 35].

R1.5 The energy column

Energy consumption results. The energy column in Table 11 reports 0.00 Wh for all methods, making it difficult to draw any comparison. Either report the values with higher precision or clarify that the measurements fall below the resolution of the measurement tool.

Neither option applies, for a reason worth stating precisely: our energy figure is a *proxy* computed as wall-clock $\times$ an assumed CPU draw, and wall-clock was not recorded for those canonical small-graph runs, so the column had no measurement behind it rather than one below resolution. Increasing the precision would therefore not have produced a comparison. We have instead made two changes that give the reviewer the comparison the comment is asking for:

1. **The energy column is removed from Table 11**, and the caption now says why, so no reader infers energy parity from a row of zeros:

“Energy is *not* reported here: wall-clock was not logged for these canonical small-graph runs, so the proxy of Section 4.5 has no input for them. We report it instead where the measurement exists, on the large-scale study (Table 10), rather than print an uninformative zero.”

2. **Energy is now reported where the measurement does exist.** The large-scale ogbn-products study logged per-run wall-clock, so Table 10 gains an **Energy (Wh)** column derived from it, and the numbers are informative: FedAvg-GAT 26.2 Wh, FairSIN 24.3 Wh, DP-FedAvg 22.6 Wh, FaVGNN 50.4 Wh, TrustFedGNN 66.2 Wh. §5.10 now reads the comparison out explicitly — roughly 40 Wh, about 16 gCO₂e, as the one-off cost of training a fair, private and robust model on a 2.4M-node graph — while labelling the figure a proxy on fixed hardware rather than a metered measurement.

No experiment was re-run for this: the wall-clock values were already in Table 10 and the energy column is an arithmetic transformation of them. We also patched the two table-generating scripts so that an unmeasured run now emits “–” instead of a spurious zero, preventing the defect from recurring.

R1.6 Text/table inconsistencies

I also noticed a few inconsistencies between the text and tables that should be corrected ...

All five were real. Each is corrected as follows.

(a) **FedFACT DPD on Bail.** The text said 0.015; Table 5 says 0.021. The table is correct and the text now reads 0.021 [§5.3, p. 19]. (The surrounding claim — that FedFACT achieves the lowest DPD among non-private methods on Bail — remains true at the corrected value.)

(b) **FairFed p -value on German.** The text quoted $p=0.006$ while Table 3 shows 0.01. The table rounds p -values to two decimals; the text now quotes the rounded value $p=0.01$, and the caption of Table 3 states the rounding convention explicitly so that text and table cannot drift again.

(c) **The AUC claim does not hold for FairGNN.** Correct: FairGNN’s $p=0.16$ is not significant. The sentence now names both the rule and the exception [§5.3, p. 18]:

“... on German its AUC is significantly below *most* non-private baselines — FedAvg-GAT, FairSIN, F²GNN ($p=0.02$) and FairFed ($p=0.01$) — the cost of DP plus the fairness/robustness constraints. The one non-private baseline it is *not* significantly below is FairGNN ($p=0.16$), whose adversarial objective is itself unstable on a graph this small (AUC 0.668 ± 0.084); we state this exception rather than generalise over it.”

(d) **The attribute-inference baseline: 0.50 versus 0.53.** The two numbers refer to two different quantities, and the figure labelled the wrong one as “chance”. Our attack samples targets *in equal numbers from each true group* (2,000 per group), so an adversary that ignores the release entirely cannot beat 0.500 on that target set — which is the baseline the table reports. The figure’s line at 0.53 is the majority-group *prevalence* in Bail, which is the chance level only for an *unbalanced* target set.

We have re-rendered Figure 3 from the already-logged attack results (no re-training; the seven (ϵ , accuracy) pairs are exactly those in Table 8, so figure and table are now guaranteed to agree). The corrected figure draws the chance line at 0.500 and labels the 0.53 prevalence line for what it is. The text [§5.6, p. 24] and both captions now explain the distinction, and the plotting script was patched so the mislabelling cannot return.

We kept 0.500 as the reference because it is both the correct comparison for a balanced target set and the more demanding one: it requires the mechanism to reduce the adversary to a coin flip rather than to a class-prior guess. Against that reference the result is stronger than we had previously stated. Each row of Table 8 is a Monte-Carlo estimate over 4,000 target draws, so its standard error is $\sqrt{0.25/4000} \approx 0.008$, and the largest excess over chance in the entire sweep is 0.013, or 1.6 standard errors. We therefore added:

“Measured against that harder reference, the residual leakage is not merely small but statistically undetectable at our sample size ... *At no privacy budget is the attack’s accuracy significantly different from 0.500* (two-sided $p \geq 0.10$ throughout). We therefore claim that FTGD reduces the differencing adversary to chance, not merely close to it — while noting that this bounds leakage only at the resolution 4,000 trials can resolve, so an advantage below roughly one percentage point would not be visible to this test.”

(e) **German DPD in the ablation versus the main table.** The ablation reports 0.045 and Table 5 reports 0.040 ± 0.035 . The reason is that the ablation is a *separate set of runs* with its own matched seed set — so that all four configurations are compared on identical seeds — whereas the main tables aggregate the full seed sets (10 on German, 5 on Bail). The ablation caption now states this explicitly and shows that all four affected cells differ by less than one standard deviation of the multi-seed mean [Table 7, p. 23].

R1.7 Scope of the DP guarantee in the abstract

Finally, the abstract and the description of the differential privacy guarantee should make it clearer that the guarantee applies to the released fairness statistics, rather than the entire model update.

Agreed. The scope was previously signalled in the abstract only by a parenthetical (“we state precisely what this guarantee covers, and what it does not, in the text”), which is too easy to skim past for a limitation this important. It is now a sentence of its own:

“We emphasise at the outset that this (ϵ, δ) guarantee covers the *released fairness statistic only*, and *not* the full transmitted model update, which retains sensitive-attribute dependence through FSER and the fairness-gradient masks; the comparison against full-gradient DP-SGD is therefore not like-for-like, and we make the exact scope precise in Section 4.2.”

Beyond the abstract, we made three further scope-tightening changes. The abstract’s closing claim and the conclusion now say “*statistic-level* sensitive-attribute DP” rather than “sensitive-attribute DP”. Contribution 2 in §1 already carried the caveat and is unchanged. Most importantly, **Table 1** previously gave TrustFedGNN a plain ✓ in the DP column, directly beside DP-FedAvg, PUFFLE and FDP-Fair, which implied guarantees of equal scope; that entry now carries a marker and the caption explains:

“[‡]Our DP entry is *not* equivalent to the record-level guarantees of DP-FedAvg, PUFFLE, FDP-Fair or FaVGNN: TrustFedGNN privatises the *released fairness statistic* rather than the whole model update (Section 4.2), so this column marks the presence of a formal (ϵ, δ) guarantee, not guarantees of equal scope.”

4. Response to Reviewer 2

We thank Reviewer 2 for identifying the two places where the paper’s framing, rather than its content, was doing readers a disservice. Both comments led to substantive rewrites.

R2.1 Practical implications of requiring the protected attribute at inference

FSER requires the protected attribute at inference time. This is currently called a “governance nuance” in one line. Given the paper’s emphasis on EU AI Act compliance, this deserves a dedicated paragraph in Section 6 if possible, by the authors to explain what this means practically for a deploying institution, such as what data access agreements or any legal obligations it creates.

We agree that this warrants a full treatment: it is the design decision in TrustFedGNN with the largest practical and legal footprint for a deployer. §6 now carries a dedicated paragraph [§6, p. 32] that first makes the technical requirement concrete — because the model attends over a neighbourhood, scoring one applicant requires the protected attribute of the *counterparties* in that applicant’s neighbourhood, not only of the applicant — and then works through four consequences for a deployer:

First, lawful basis. Gender, race and ethnicity are special-category data under GDPR Art. 9; processing them in an automated decision requires an Art. 9(2) condition, and in several member states the substantial-public-interest route additionally needs a national-law basis. The EU AI Act anticipates this tension: Art. 10(5) permits processing special-category data *for the purpose of bias detection and correction* in high-risk systems, subject to safeguards including technical limits on re-use,

pseudonymisation, and deletion once bias correction is complete. FSER is squarely a bias-correction mechanism and so is the kind of processing that provision contemplates — but an institution must document that reliance rather than assume it, and Art. 10(5) is a permission to process, not a waiver of Art. 9 or of a DPIA.

Second, data-access agreements. In the cross-silo setting the counterparties in a neighbourhood may be another institution’s customers, so inference-time access to s can require inter-institutional agreements that plain FedAvg does not — a governance cost that partly offsets the “no raw data leaves the silo” benefit of federation.

Third, collection risk. Where s is not already held, obtaining it means either new collection (consent, minimisation, retention limits) or *inference* of a protected attribute from proxies, which is itself a discriminatory-profiling risk and which we do not recommend.

Fourth, purpose limitation. s must reach the attention mechanism without reaching the classifier as a feature: our implementation keeps it out of $\mathbf{X}$, but that separation is an engineering invariant a deployer must verify and an auditor must test, not a property the architecture enforces by itself.

The paragraph closes by telling a reader who *cannot* meet these obligations what to do: FSER is then the wrong component, and the ablation (Table 7) shows FTGD and BFWA operating without it at a quantified fairness cost. We also added a new limitation item (12) recording that “until this is done, TrustFedGNN is only deployable where such processing is lawful and governed”, and a future-work direction on removing the dependence altogether — by distilling FSER into an s -free student model, or by driving the edge correction from the DP-noised statistics instead of a hard group indicator.

R2.2 Making the trust-score limitation more prominent

Also, the paper mentions the composite trust score is not decisive evidence, but this appears after the score is already presented. If possible, highlight this limitation more prominently, which can be helpful for readers.

We agree, and have inverted the order so the caveat precedes the number rather than following it. The subsection is retitled “**Composite trust score: read this caveat first**” and now opens with the limitation, before the score is defined [§6, p. 31]:

“Because a single headline number invites exactly the over-reading we want to avoid, we state its limits *before* presenting it rather than after. The composite score below is a *communication tool* ... It is not a certification of trustworthiness, not a substitute for the per-axis evidence, and not decisive evidence of superiority. Its ranking depends on choices we make — the reference constants, the unit weights, and the power p — and a reader who chooses different constants may obtain a different ordering. Auditors, deployers and reviewers should therefore treat the per-axis results (Tables 4–11) as the primary empirical findings, and the composite as a summary of them. With that framing fixed, the construction is as follows.”

The corresponding limitation item was also strengthened, and now adds a point we had not made anywhere: the composite covers only the axes computable from a given run, so the three-axis version in Table 12 omits calibration and robustness entirely, and “a high score is a statement about the axes included, not about the system as a whole” [§7, Limitation (10), p. 35].

5. Closing

We believe the manuscript is considerably stronger for these comments. The technical content is unchanged — no result has been altered and no new experiment was required — but the claims are now precisely bounded by the evidence, the text and the tables agree throughout, the scope of the privacy guarantee is unambiguous, and the two governance questions the Editor

and Reviewer 2 raised (inference-time access to protected attributes, and the standing of the composite trust score) are treated at the length they deserve.

We would like to thank the Reviewers once more for the care they invested, and the Editor for the opportunity to revise.

Sincerely,

Minh Ngoc Dinh, on behalf of all authors

Corresponding author